1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L19 – DFD+ERD**

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3 **ER Diagram**

4 **Representing Data Stores**

- In DFD, we have seen data stores
- So far, used only labels to represent those
- Labels don't reveal "structure" and "organization" of data store
- Also there may be relationships between various data elements, not revealed by simple labels

5 **Representing Data Stores**

- One way to "express" rich internal structure, organization and relationships in data stores is to use ER (stands for Entity-Relationship) diagrams

6 **E-R Diagram**

- Proposed by Peter Chen (1976)
- Includes entities and relationships
- Most common representation for relational databases

7 **Basic Components**

8 **Modeling**

- A *data store (database)* can be modeled as:
  - A collection of entities
  - Relationship among entities

9 **Entity**

- An object that exists and is distinguishable from other objects (e.g., person, company, student, customer)
- Have *attributes* (e.g., person have *names* and *addresses*)
- Entity set - set of same type entities that share same properties
  - E.g. set of all persons
- Each entity must be uniquely identifiable

10 **Attributes**

- Descriptive properties of entity
- A particular instance of an attribute is a value
- Domain- set of permitted values of an attribute

11 **Attribute Types**

- Simple - contains only atomic values
- Multi-valued - contains several atomic values

12 **Attribute Types**

- Null attributes - used when an entity does not have a value for an attribute
- Derived attribute - value derived from other attributes or entities
  - E.g., Age from DOB

13 **ERD with Multivalued and Derived Attributes**

14 **Relationship**

- An association between entities

  Example:

  Sam *depositor* E-100

  customer entity relationship *account* entity

15 **Relationship Degree**

- Number of entities in a relationship
  - Binary (or degree 2) – two entities (most common)
  - May be more than two (ternary relationship) – although rare

16 **Example – Ternary Relation**

17 **Relationship Types**

- One-to-one
- One-to-many
- Many-to-one
- Many-to-many

18 **One – To - One Relationship**

- A customer entity is associated with at most one (possibly 0) loan via borrower
- A loan entity is associated with at most one (possibly 0) customer via borrower

19 **One-To-Many Relationship**

- A loan entity is associated with at most one customer via *borrower*
- A customer entity is associated with several (including 0) loans via *borrower*

20 **Many-To-One Relationships**

- A loan entity is associated with several (including 0) customers via *borrower*
- A customer entity is associated with at most one loan via *borrower*

21 **Many-To-Many Relationship**

- A customer entity is associated with several (possibly 0) loans via borrower
- A loan entity is associated with several (possibly 0) customers via borrower

22 **Alternative Notation for Relation**

- (min..max) notation
  - Min indicates each entity is in relationship at least min times
  - Max indicates each entity is in relationship at most max times
- Special cases
  - min=0 - does not have to be in relationship (optional)
  - max=* - may be in arbitrarily many times

23 **E-R Diagram with Alternative Notation**

24 **Note**

- We covered the very basics
- ERD is more expressive, with a rich set of notations, covering many more aspects

25 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 6
- Roger S Pressman (2020). Software Engineering: A Practitioner's Approach, 9th ed, McGraw-Hill Education, New York